

Multi-Task Generative Restoration and Style-Conditioned Image Translation: From Denoising Autoencoders to Soft Mixture-of-Experts and Conditional GANs

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Abstract

This research technical report investigates four interconnected generative paradigms spanning image degradation recovery, adaptive specialist routing, differentiable soft mixture-of-experts (MoE), and style-conditioned paired image translation. Using the Oxford-IIIT Pet benchmark, we first formulate a universal multi-corruption convolutional denoising autoencoder capable of reconstructing high-fidelity targets from heterogeneous inputs (clean, salt-and-pepper noise, Gaussian blur, and rectangular occlusion). Second, we propose a modular hard-routing architecture that classifies degradation regimes and dispatches inputs to isolated specialist autoencoders with an identity bypass for clean signals. Third, to mitigate hard routing boundaries and ambiguity, we establish a jointly trained differentiable soft mixture-of-experts model featuring entropy-balanced gating. Fourth, we implement a conditional generative adversarial network (cGAN) based on a U-Net generator and PatchGAN discriminator trained on the FS2K facial sketch benchmark under categorical style embeddings. All production models are exported and validated via ONNX runtime and served through a containerized FastAPI and React/Tailwind web platform orchestrated via Docker Compose.

Keywords: Denoising Autoencoders, Mixture-of-Experts, Conditional GAN, PatchGAN, Hard-Routing, Image Restoration, ONNX, Optuna.

I. INTRODUCTION

Modern computer vision pipelines deployed in real-world environments frequently encounter stochastic signal corruptions, ranging from sensor noise and optical defocusing to physical occlusions. Traditional restoration approaches predominantly focus on single-corruption specialists or rigid monolithic architectures.

This paper investigates the continuum between monolithic universal reconstruction, discrete hard-routed specialist dispatching, and continuous differentiable soft mixtures-of-experts. Furthermore, we explore paired cross-domain image synthesis by formulating a style-conditioned Generative Adversarial Network for facial sketch generation.

II. TASK 1: UNIVERSAL DENOISING AUTOENCODER

A. Architecture and Bottleneck Design

The Universal Autoencoder comprises a symmetrical convolutional encoder-decoder U-Net topology processing 128x128x3 RGB representations. The encoder progressively compresses spatial dimensions across four downsampling stages with channel progression 3->C->2C->4C->8C (C=32). A convolutional spatial bottleneck regularizes feature representations with 2D spatial dropout. Multi-scale skip connections link corresponding encoder feature maps directly to decoder upsampling stages at 64x64, 32x32, and 16x16 resolutions, enabling high-frequency texture and contour preservation without blurring. The decoder mirrors this pathway via transpose convolutions terminated by a Sigmoid activation.

B. Composite Loss Formulation

To preserve both pixel-accurate intensity and high-frequency structural contours, the network is optimized using a weighted linear combination of L1 reconstruction loss and Structural Similarity Index Measure (SSIM):

$$L_{\text{universal}} = \alpha * L1(x, x_{\text{hat}}) + (1 - \alpha) * (1 - \text{SSIM}(x, x_{\text{hat}}))$$

where α in $[0,1]$ balances low-frequency fidelity and edge sharpness. Hyperparameter exploration via Optuna converged on $\alpha=0.82$, learning rate $\eta=1.0e-3$, and base channel capacity of 32. The model achieved a best validation loss of 0.0604.

III. TASK 2: HARD-ROUTED SPECIALIST AUTOENCODERS

A. Corruption Classification

Task 2 decomposes the problem into a classification gate followed by specialist restorers. A 4-stage convolutional classifier predicts class probabilities across four balanced degradation categories. The classifier achieved a best validation accuracy of 94.70% with macro F1-score of 0.9471.

B. Routing Mechanism and Identity Bypass

Inference executes conditionally according to the predicted class. Clean inputs use identity bypass. Salt-and-Pepper, Gaussian Blur, or Occlusion predictions route to corresponding specialist autoencoders.

C. Specialist Training

Each specialist shares the U-Net architecture but is warm-started from universal autoencoder weights and fine-tuned on its target corruption type. Validation losses: Salt-and-Pepper: 0.0444, Blur: 0.0449, Occlusion: 0.0728.

IV. TASK 3: JOINTLY TRAINED SOFT MIXTURE-OF-EXPERTS

A. Differentiable Gating Formulation

Hard routing incurs vulnerability when corruptions co-occur or border decision boundaries. Task 3 resolves this with a differentiable Soft MoE where the gating network produces softmax-normalized routing weights with temperature scaling.

B. Joint Optimization Objective

The joint loss regularizes reconstruction accuracy, classification grounding, and routing entropy with a balance term preventing expert starvation. The model achieved best validation loss of 0.0557.

V. TASK 4: CONDITIONAL GAN FOR FACE-TO-SKETCH

A. Generator and PatchGAN Discriminator

Task 4 synthesizes facial sketches from photographs using the FS2K benchmark dataset (2,104 pairs, 3 styles). The generator uses an 8-stage U-Net with style embeddings spatially tiled and concatenated with input. The discriminator implements a 70x70 PatchGAN.

B. Ink-Weighted Loss Formulation

To prevent washed-out sketches, we employ an ink-weighted L1 loss with 5x penalty on dark pencil strokes, plus edge gradient loss for crisp contours. Best validation L1: 0.2387.

VI. EXPERIMENTAL RESULTS

Summary of training results across all models:

Model	Metric	Value
Universal AE (U-Net)	Val Loss	0.0604
Corruption Classifier	Val Accuracy	94.70%
Corruption Classifier	Macro F1	0.9471
Salt-Pepper Specialist	Val Loss	0.0444
Blur Specialist	Val Loss	0.0449
Occlusion Specialist	Val Loss	0.0728
Soft MoE (Joint)	Val Loss	0.0557
Face-to-Sketch cGAN	Val L1	0.2387

All seven trained models were exported to ONNX format with strict numerical parity verified (max error < 2.5e-5).

VII. APPLICATION ARCHITECTURE AND DEPLOYMENT

The system is implemented as a unified full-stack application. The backend uses FastAPI with ONNX Runtime (<25ms inference). The frontend uses React 19 and Tailwind CSS 4. Post-processing includes Lanczos upsampling, unsharp mask sharpening, and autocontrast normalization. The entire ecosystem is containerized via Docker Compose.

VIII. CONCLUSION

Universal autoencoders deliver generalist resilience (val loss 0.0604). Specialist routing achieves superior per-corruption reconstruction (0.0444-0.0728). Soft MoE improves further (0.0557) via differentiable routing and entropy-balanced gating. The conditional GAN successfully maps facial topography onto distinct artistic sketch domains. Future work includes extending the corruption taxonomy and exploring attention-based gating mechanisms.

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